Technical Document #361: Software Engineering - Comprehensive Overview

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Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Continuous deployment (CD) automatically deploys code changes to production.

- Continuous integration (CI) automatically builds and tests code changes.

- Microservices architecture structures applications as independently deployable services.